





Relational Algebra

Basic Operations & Query Processing Concepts



Learning Objectives

By the end of this lecture, you should be able to...

01

Explain the purpose of Relational Algebra as a formal, procedural way to query relations.

02

Distinguish Selection and Projection , rows versus columns, and recognize their notation.

03

Apply Selection, Projection, Rename, Union, Difference, and Cartesian Product to simple queries.

04

Combine relational algebra operations to express more meaningful database questions.

05

Connect Relational Algebra with SQL and the logical query-processing work of a DBMS.



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Relational Algebra

Basic Operations & Query Processing Concepts

How do we ask a database a precise question?

QUESTION

“Show me all CSE students.”



RELATIONAL ALGEBRA

$\sigma_{\text{Department}='CSE'}(\text{Student})$



RESULT

Only the required rows



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From a Question to a Database Result

Human question



Operation



Result relation

QUESTION

Find all CSE students.

$\sigma_{\text{Department}='CSE'}(\text{Student})$

FILTER THE ROWS

σ

RESULT

Rahul • CSE
Raj • CSE

TABLE IN → OPERATION → TABLE OUT



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Why Do We Need Relational Algebra?

A DBMS needs a formal way to express data requests

CSE students

Find all students from CSE.

DBMS learners

Display students enrolled in DBMS.

Salary filter

Find employees earning above ₹50,000.

Teaching list

List courses taught by a professor.

SOLUTION

Relational Algebra — a mathematical framework for querying relations.



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What Exactly Is Relational Algebra?

Definition → characteristics → example

DEFINITION

A procedural query language consisting of operations used to manipulate and retrieve data from relational databases.

KEY CHARACTERISTICS

- Built on relational theory
- Relations as input
- Relation as output
- Mathematical operations
- Foundation for SQL query processing

EXAMPLE

Student relation

↓

$\sigma_{\text{Department}='CSE'}$

↓

New relation

Input → Operation → Output



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Procedural vs Declarative

Think: HOW vs WHAT

PROCEDURAL — RELATIONAL ALGEBRA

Specifies HOW

User describes operations

Sequence/composition matters

Example:

$\sigma_{\text{Department}='CSE'}(\text{Student})$

DECLARATIVE — SQL

Specifies WHAT

User describes the required data

DBMS chooses how to retrieve it

Example:

```
SELECT * FROM Student WHERE  
Department='CSE';
```

HOW ↔ WHAT



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The Relational Algebra Toolkit

Classify operations by the number of input relations

UNARY — ONE RELATION

σ Selection
 π Projection
 ρ Rename

BINARY — TWO RELATIONS

$\cup$ Union
– Difference
 $\times$ Cartesian Product

ADVANCED

$\bowtie$ Join $\div$ Division

Question to ask: “How many tables do I need as input?”



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Selection Operation (σ)

Selection answers: “Which ROWS do I want?”

RollNo	Name	Dept	Age
101	Rahul	CSE	21
102	Anita	ECE	22
103	Raj	CSE	19
104	Priya	ME	23

QUERY

Find CSE students

EXPRESSION

$\sigma_{\text{Department}='CSE'}(\text{Student})$

RESULT

101	Rahul	CSE	21
103	Raj	CSE	19

$\sigma \rightarrow$ ROW FILTER



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Selection Conditions

Make the filter more expressive

COMPARISON

= ≠ < ≤ > ≥

LOGIC

AND ∧

OR ∨

NOT ¬

CHALLENGE

Find CSE students
older than 20.

$\sigma_{\text{Department}='CSE' \wedge \text{Age}>20}(\text{Student})$

Result → Rahul only

Think of selection as a WHERE-like filter on rows.



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Projection Operation (π)

Projection answers: "Which COLUMNS do I want?"

RollNo	Name	Dept	Age
101	Rahul	CSE	21
102	Anita	ECE	22
103	Raj	CSE	19
104	Priya	ME	23

QUERY

Display only student names

EXPRESSION

$\pi_{\text{Name}}(\text{Student})$

RESULT

Name
Rahul
Anita
Raj

$\pi \rightarrow$ COLUMN FILTER



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Selection vs Projection

The distinction students must remember

SELECTION σ

Selects ROWS

Horizontal filtering

Uses CONDITIONS

Example:

$\sigma_{\text{Department}='CSE'}(\text{Student})$

PROJECTION π

Selects COLUMNS

Vertical filtering

Uses ATTRIBUTE LIST

Example:

$\pi_{\text{Name}}(\text{Student})$

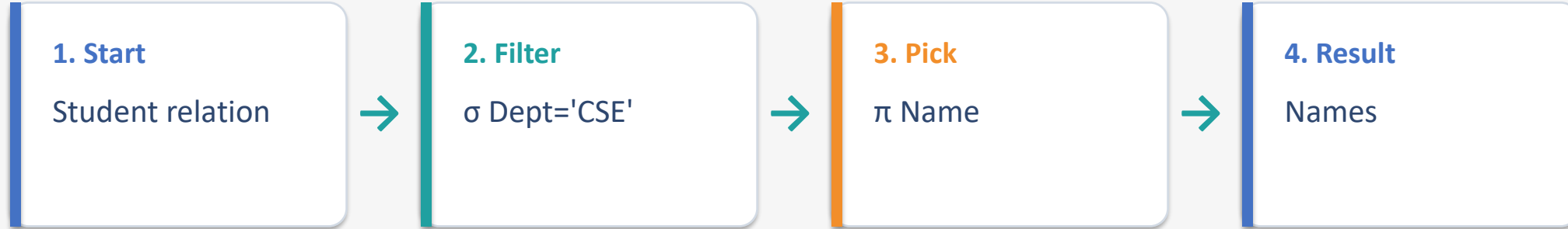
$\sigma \rightarrow \text{ROWS}$ | $\pi \rightarrow \text{COLUMNS}$



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Combining Operations: Build a Query Pipeline

Complex questions can be solved with simple operations



FINAL EXPRESSION

$$\pi_{\text{Name}}(\sigma_{\text{Department}='CSE'}(\text{Student}))$$

Question → sequence of operations → final relation



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Rename Operation (ρ)

Change the reference name without changing the data

BEFORE

Student

The original relation name



AFTER

$\rho_{\text{Student_Info}}(\text{Student})$

Now referred to as Student_Info

WHY RENAME?

Same relation used multiple times • clearer expressions • useful with joins



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Union Operation (U)

Combine tuples from two compatible relations

CSE_Students

101

102

U

ECE_Students

103

104

=

RESULT

101

102

103

104

CONDITION

Same number of attributes + compatible domains



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Set Difference (–)

Find tuples present in one relation but not the other

ALL_STUDENTS

101
102
103

–

GRADUATED_STUDENTS

101

=

RESULT

102
103

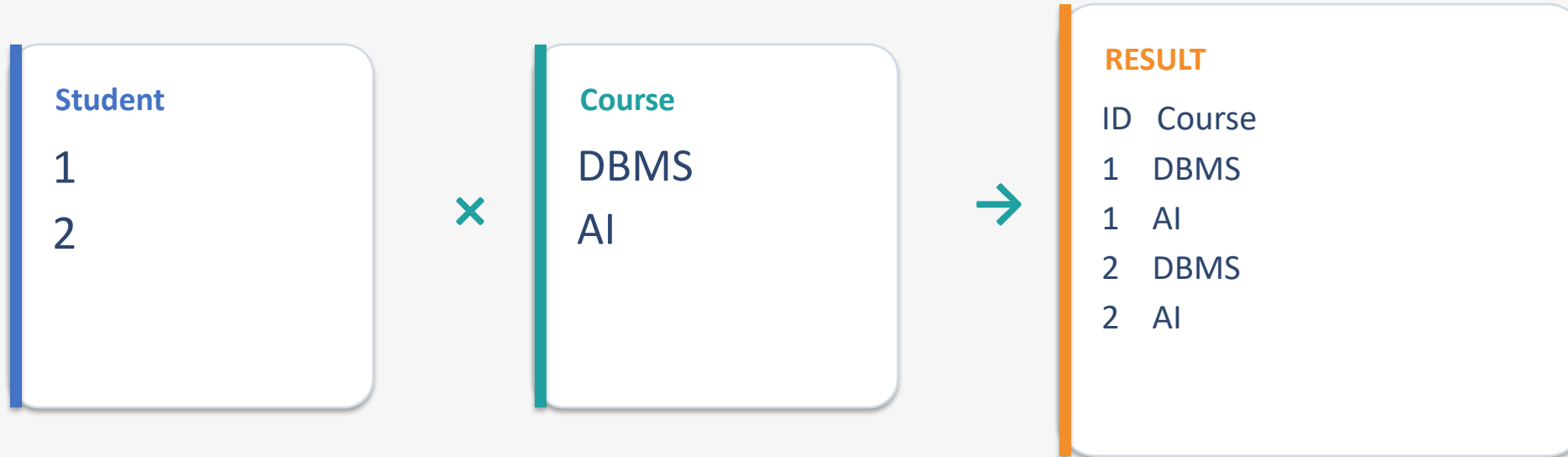
Practical question: “Who has NOT graduated?”



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Cartesian Product (×)

Everyone meets everyone



2 students × 2 courses = 4 combinations



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Cartesian Product Explosion 🤯

Mathematically simple — potentially huge in practice



STUDENTS

10,000

×

COURSES

500

=

POSSIBLE COMBINATIONS

5,000,000

PROBLEM

Most combinations may be meaningless.

SOLUTION

Use JOIN to combine related records.



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Which Operation Would You Use?

Read the question → identify the operation

Find CSE students

Selection

Display student names

Projection

Combine CSE + ECE

Union

Find students who have not paid fees

Difference

Generate every student-course combination

Cartesian Product



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Relational Algebra → SQL

Same logical ideas, practical query language

SELECTION

$\sigma_{\text{Age} > 20}(\text{Student})$

SQL:

```
SELECT * FROM Student WHERE Age > 20;
```

PROJECTION

$\pi_{\text{Name}}(\text{Student})$

SQL:

```
SELECT Name FROM Student;
```

KEY IDEA

Relational algebra gives the mathematical foundation; SQL provides the practical interface.

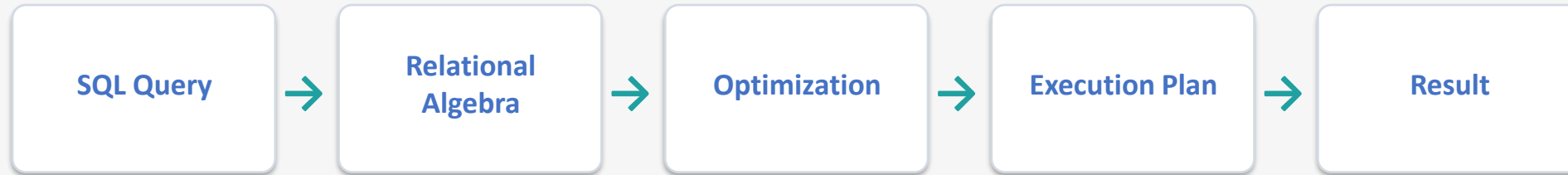
SQL hides much of the mathematical complexity from users.



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What Happens Inside a DBMS?

A conceptual query-processing pipeline



WHY OPTIMIZE?

Less unnecessary work • smaller intermediate results • better execution



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Relational Algebra Is Still Relevant

Same data-processing ideas, modern platforms

Spark DataFrames

Filtering + transformation

Distributed SQL

Logical query processing

Cloud databases

Query optimization

NoSQL systems

Similar filtering/transformation ideas

CORE IDEA

The tools change. The core idea of transforming data remains.



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Summary

What you should remember

✓ Relational algebra is a procedural way to express database queries.

✓ Projection (π) → chooses COLUMNS using an attribute list.

✓ Union ($\cup$) → combines union-compatible relations.

✓ Cartesian product ($\times$) → forms all possible combinations.

✓ Selection (σ) → chooses ROWS using a condition.

✓ Rename (ρ) → changes relation or attribute names.

✓ Difference ($-$) → returns tuples in the first relation but not the second.

✓ Complex queries are built by combining these operations.



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Learning Reflections

Explain the idea in your own words

1 What is the difference between selection (σ) and projection (π)? Give one example of each.

2 How would you express “names of CSE students” using relational algebra?

3 Why must the two relations in a UNION operation be union-compatible?

4 Why can a Cartesian product become very large, and why are JOIN operations useful?



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Homework

Explain how relational algebra provides a procedural and mathematical foundation for querying relational databases, and describe how selection, projection, rename, union, set difference, and Cartesian product operations are used to retrieve and manipulate relations. Explain the notation, purpose, input and output of each operation.



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Thanks!



Heads Up!

These slides are just a **guide** — not the full story!

To really learn , make sure to read your book, and explore other resources too.